THE USE OF HIGH-PERFORMANCE POLYMERS IN DENTISTRY

YÜKSEK PERFORMANSLI POLIMERLERIN DIŞ HEKIMLIĞINDE KULLANIMI

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ABSTRACT

Various materials, such as zirconia, ceramics, titanium, and Co-Cr alloys, are used as veneer and framework materials, especially in prosthetic dental treatments. These materials have their advantages and disadvantages. In recent years, high-performance polymers have been introduced as alternative dental framework materials due to their high fracture resistance, biocompatibility, lightweight, and shock absorption properties. Poliarileterketons, a type of high-performance polymer, belong to the group of thermoplastic polymers. Among the poliarileterketon family used in dentistry, polyether ether ketone (PEEK) and polyether ketone ketone (PEKK) are organic thermoplastics. This review aims to examine, evaluate, and review the usage areas of PEEK and PEKK materials in dentistry in detail. For this purpose, articles published between 2010 and 2023 related to "usage areas of PEEK and PEKK materials in dentistry" were examined in PubMed/MEDLINE, Web of Science, Google Scholar, Cochrane Library, and Scopus databases. The reviewed studies emphasise that PEEK and PEKK materials possess suitable physical properties and can be used as infrastructure materials by bonding them to upper structure materials.

Keywords: high-performance polymers, Peek, Pekk, dental framework material, prosthetic dental treatment

ÖZET

Diş hekimliği alanında, özellikle protetik diş tedavisinde, çeşitli üst ve alt yapı materyalleri olarak zirkonya, seramik, titanyum ve Cr-Co gibi malzemeler kullanılmaktadır. Bu materyallerin kendi arasında avantajlar ve dezavantajlar bulunmaktadır. Son yıllarda, diş hekimliği uygulamalarında yüksek kırılma dayanımı, biyouyumluluk, hafiflik ve şok emme gibi özelliklere sahip olmaları nedeniyle, yüksek performanslı polimerler alternatif dental alt yapı malzemeleri olarak kullanılmaya başlanmıştır. Poliarileterketonlar, yüksek performanslı polimerlerden biridir ve termoplastik polimerlerdir. Diş hekimliğinde kullanılan poliarileterketon ailesinin üyeleri polietereterketon (PEEK) ve polieterketonketon (PEKK) organik termoplastiklerdir. Bu derlemenin amacı, PEEK ve PEKK malzemelerinin diş hekimliğindeki kullanım alanlarını ayrıntılı bir şekilde incelemek, değerlendirmek ve bu konuda yapılan çalışmaları gözden geçirmektir. Bu amaçla, 2010-2023 tarihleri arasında PubMed/MEDLINE, Web of Science, Google Scholar, Cochrane Library ve Scopus veri tabanlarında "PEEK ve PEKK malzemelerinin diş hekimliğinde kullanım alanları" ile ilgili yayınlanmış makaleler incelenmiştir. İncelenen çalışmalar, PEEK ve PEKK malzemelerinin

uygun fiziksel özelliklere sahip olduğunu ve üst yapı materyallerine bağlanarak alt yapı malzemesi olarak kullanılabileceğini vurgulamaktadır.

Anahtar Kelimeler: yüksek performaslı polimerler, PEEK, PEKK, dental altyapı materyali, protetik diş tedavisi

INTRODUCTION

In dentistry, various restoration options are available to treat functional and aesthetic deficiencies caused by tooth and soft tissue loss. Some of these prosthetic restoration options used to restore function and aesthetics include Laminate Restorations, crowns, bridges, removable partial dentures, complete dentures, and fixed/removable prostheses on implants (Saeed, F. et al., 2020; Tunçdemir, A., & Özcan, E., 2015). These prosthetic restorations are constructed using both framework and veneer materials. Selecting the appropriate material for the restoration and case is of great importance for the success and longevity of the prosthetic restoration. The essential characteristics a material should possess for its use in dentistry are aesthetics, durability, resistance to wear, non-corrosiveness, low thermal conductivity, high biocompatibility, and the ability to mimic natural teeth and surrounding tissues structurally (Çelik, H., et al., 2021; Saeed, F., et al., 2020; Nallaswamy, D., 2017). A balance should be struck among these properties when choosing materials for use in dentistry. If a material possesses excellent aesthetic properties and high durability but is not tissue-friendly, it may not be suitable for use in dentistry. For fixed or removable prostheses, options for framework materials include metal alloys, titanium, and zirconia (Çelik, H., et al., 2021). For many years, crowns, bridges, and removable dentures made with metal alloys in dentistry have involved restorations with multiple layers in concealing the metal framework. However, the available grey colour of metal cannot be fully compensated. As a result, alternative materials have gradually replaced restorations with metal frameworks over the years. Esthetic restorations that have been under development in dentistry for a long time have gained a different identity with the emergence of zirconia. The use of zirconia has become widespread due to its hardness, aesthetic appeal, and biocompatibility. However, the disadvantage of zirconia is its high cost compared to chrome-cobalt-based prosthetic restorations (Sereno, N., 2015; YAĞCI, Ö., 2022; Webber, B., McDonald, A., and Knowles, J., 2003). The inability of metal frameworks to provide the desired aesthetics and the high cost of zirconia has led to the search for different materials in dentistry. As a result, research in high-performance polymers has gained momentum (Kewekordes, T., Wille, S., & Kern, M., 2018). With technological advancements, applications using conventional methods are replaced by PEEK and PEKK materials. As companies have also made advancements in this field, milling machines have been developed for the production of crown and bridge restorations, and high-performance polymer blocks have been produced (Kewekordes, T., Wille, S., & Kern, M., 2018; Dawson, J. H., 2018; Ergün, G., & Tekli, B., 2022). This review aims to examine and evaluate the physical and chemical properties, advantages and uses in dentistry of PEEK and PEKK materials, which are high-performance polymers, and to review the studies on this subject.

METHOD

This article is designed as a review article supported by a literature review in various scientific databases. A search was carried out on the databases PubMed / MEDLINE, Web of Science, Google Scholar, and Scopus to find articles published between 2010 and 2023 on High-Performance Polymers in Dentistry and aimed at review purposes. Clinical trials randomized controlled trials, prospective and retrospective cohort studies, systematic reviews and meta-analyses of "PEEK and PEKK" were primarily examined.

RESULTS AND DISCUSSION

High-performance polymers

High-performance polymers, known as Polyaryletherketones (PAEKs), are a group of semi-crystalline thermoplastic polymers that vary in ketone and ether ratios (Fuhrmann et al., 2014; Ergün, G., & Tekli, B., 2022). They possess an aromatic chain with a chemical structure that remains stable at high temperatures and under mechanical forces. Semi-crystalline aromatic polymers exhibit excellent impact resistance, chemical resistance, high mechanical fatigue strength, a remarkably low leakage and fluid absorption tendency, and good wear properties (Kurtz et al., 2019; McKeen, L., 2019). The PAEK polymer structures are nearly identical—those with higher ketone content exhibit increased chain polarity and hardness. Polyetheretherketone (PEEK) and Polyetherketoneketone (PEKK) are organic thermoplastic polymers in the PAEK family used in dentistry. High-performance polymers have been presented as an alternative material in dentistry due to their sufficient fracture resistance, close-to-ideal stress distribution, and shock absorption properties (Fuhrmann et al., 2014; Bathala et al., S., 2019; McKeen, 2019). PEEK and PEKK differ in the ketone/ether linkage ratio. PEEK has a ketone/ether linkage ratio of 33%, while PEKK has a ratio of 67%. This ratio leads to thermal and mechanical differences between PEEK and PEKK. PEKK exhibits better properties than PEEK in terms of the glass transition temperature, melting temperature, tensile strength, flexural strength, and compressive strength (Bathala et al., 2019; McKeen, 2019; Ergün, G., & Tekli, B., 2022). (Table 1)

Table 1. Properties of PEEK and PEKK

Property	PEEK	PEKK
Glass transition temperature (Tg) °C	143	165
Melting Temperature (Tm) °C	334	386
Tensile strength (MPa)	93	110
Flexural strength (MPa)	170	193
Compressive strength (MPa)	117	206

Polyetheretherketone (PEEK)

Polyetheretherketone (PEEK) is a thermoplastic polymer used for years in industrial, engineering, and medical applications. In recent years, PEEK, a high-performance polymer, has been introduced to the market in dentistry (Rocha et al., 2016). It has been utilized as a framework material for implant applications, fixed and removable prosthesis restorations, an alternative post-core material, and also in the production of clasps and restorative implant components (Skirbutis et al., 2018; Rocha et al., 2016; Sammany, Sultan, & Ide, 2019). Modified PEEK material containing 20% ceramic fillers exhibits good mechanical properties and superior biocompatibility. It can be used for prosthetic production with CAD/CAM procedures (Benli et al., 2020). The main advantages of this modified PEEK material include heat resistance, good wear resistance, and elastic moduli of 4-5 GPa (providing elasticity similar to bone and dentin). These properties allow it to act as a stress absorber, reducing the forces transferred to the restoration and dental root. PEEK has high fracture resistance (1383 N). It has a tensile strength of 93 MPa, flexural strength of 170 MPa, and compressive strength of 117 MPa while being lightweight. It is chemically inert and is shown as an alternative to glass ceramics and metals due to its shock-absorbing property (Skirbutis et al., 2018). It does not cause allergic reactions and can be polished well. Additionally, its low plaque affinity prevents bacterial plaque formation (Bathala et al., 2019). PEEK is preferred in lightweight prosthetic applications that do not contain metals (Güven et al., 2020). (Table 2)

Table 2. Properties of PEEK

Property	
Modulus of Elastic (GPa)	8,3
Bending Strength (MPa)	140
Tensile Strength (MPa)	115
Toughness (MPa)	210
Glass transition temperature (Tg) °C	143
Melting Temperature (Tm) °C	117

Polyetherketoneketone (PEKK)

PEKK offers superior mechanical and physical properties to PEEK due to its second ketone group, including better polishability, 80% higher compressive strength, low plaque affinity, and a bone-like elastic modulus (Lee et al., 2017). As a member of the same family, it exhibits improved physical characteristics compared to PEEK. PEKK has a tensile strength of 110 MPa, compressive strength of 206 MPa, and flexural strength of 193 MPa (Fuhrmann et al., 2014). To enhance its mechanical properties and achieve a more aesthetic colour, titanium dioxide (TiO2) has been added to the chemical structure of PEKK. This results in an ivory-coloured and opaque material, making it more useful for various applications. Furthermore, due to their acceptable fracture resistance, better stress distribution, and shock absorption properties, high-performance polymers such as PEKK provide an alternative to metal-supported ceramics (Han et al., 2016). The manufacturer reports that PEKK has a compressive strength (246 MPa) similar to dentin (297 MPa) but a lower elastic modulus (5.1 GPa) (Powers, Sakaguchi, & Craig, 2012). It can be used for prosthetic production with CAD/CAM procedures. It is mechanical superiority, and suitable biocompatibility make it suitable as a framework material for fixed, removable, and implant-supported prostheses. PEKK material is also considered a potential post-core material due to its stress distribution and shock absorption properties (Alqahtani, 2021; Lee et al., 2017; Song et al., 2018). (Table 3)

Table 3. Properties of PEEK

Property	
Modulus of Elastic (GPa)	5,1
Bending Strength (MPa)	200
Compressive Strength (MPa)	246
Tensile Strength (MPa)	115
Toughness (MPa)	252
Glass transition temperature (Tg) °C	163
Melting Temperature (Tm) °C	363

Manufacturing Techniques for PEEK and PEKK

Various manufacturing techniques are employed to fabricate dental restorations and devices using PEEK and PEKK materials. Injection molding, machining, milling (CAD/CAM), and additive manufacturing (3D printing) are the primary techniques utilised in dentistry. Recent technological advancements have allowed to produce of complex and customised dental components with enhanced accuracy and efficiency.

The Use of High-Performance Polymers in Dentistry

Applications of PEEK and PEKK in Prosthetics

PEEK and PEKK materials find applications in fabricating dental crowns, bridges, veneers, inlays/onlays, and endodontic posts. These materials offer excellent mechanical properties, wear resistance, and biocompatibility, allowing for durable and esthetic restorations. Recent research studies have focused on optimizing fabrication techniques and investigating the clinical performance of PEEK and PEKK restorations.

PEEK and PEKK materials are used in removable partial dentures and base materials. They offer improved esthetics, strength, and flexibility compared to traditional materials, providing patients with more comfortable and functional prosthetic solutions. Recent research studies have evaluated the performance and patient satisfaction of PEEK and PEKK-based prosthetics.

In a study evaluating the optical properties of PEKK compared to metal and zirconia in prosthetic restoration applications, it was found that PEKK does not have a significant difference in optical properties compared to these materials. In another study, PEKK was compared to IPS e.max press, Lava Ultimate, and Vita Enamic regarding flexural strength. The results showed that PEKK has sufficient mechanical and physical properties to be used as a permanent monolithic crown. However, despite these positive characteristics, PEKK and PEEK have low translucency and greyish colour, limiting their use as monolithic materials in dental restorations. It has been reported that PEEK, with its low elastic modulus, offers advantages over metal-supported ceramics in reducing occlusal forces and decreasing debonding rates when used as a veneering material with indirect light-polymerized composite resin (Alqurashi et al., 2021).

It has been reported that CAD/CAM-milled three-unit fixed prostheses with PEEK frameworks have a fracture strength of 2055 N. These values are higher than the specified values for lithium disilicate glass-ceramic (950 N), In-Ceram Alumina (851 N), In-Ceram Zirconia (841 N), and zirconium (981-1331 N) (Stawarczyk et al., 2015). In a study by Stawarczyk et al., 2013, they examined veneerless three-unit frameworks with a connector diameter of 7.4 mm made from PEEK material. They reported deformation at a compressive force of 1200 N and fracture in the connector at 1385 N. Therefore, PEEK has been stated to be a suitable framework material for fixed prostheses. In a study by Cekic-Nagas et al., 2016 evaluating the fracture strength of different inlay materials, CAD-CAM fibre block and PEEK-based inlay-retained bridges showed high fracture strength. In a study by Tartuk et al., 2016 comparing the compressive strengths of crowns made from PEEK, hybrid ceramic, and zirconia under axial forces, zirconia crowns exhibited higher compressive strength than hybrid ceramic and PEEK crowns.

Costa-Palau et al. reported in clinical reports that when they used PEEK as an alternative to conventional materials and methods to fabricate a maxillary obturator for a patient with an oronasal defect, the obturator demonstrated good retention, easy polishability, lightweight, and biocompatibility.

Post and core systems aim to increase root resistance by placing a foundation inside the root canal (Ordu et al., 2017). PEEK and PEKK are also used as post and core materials due to their ability to distribute forces along the root and their ability to be produced by the shape of the root canal, thanks to their close elastic modulus to dentin (Ordu et al., 2017; Oyar et al., 2017; Bathala et al., 2019; Alqurashi et al., 2021). It has been found that PEKK post and core systems exhibit an equal distribution on the intraarticular surfaces. The probability of root fracture is shown to be lower compared to traditional post and core systems. When examining the conducted studies, it is emphasized that PEKK can be an alternative to PEEK, metal, and fiber-based post and core systems, thanks to its shock absorption ability and appropriate mechanical strength. PEKK has higher fracture resistance than metal and fiber posts (Alqurashi et al., 2021). In a study evaluating the fracture strength of PEEK, zirconium oxide,

and glass fiber post and cores, Özarslan et al. (2021) reported that fractures occurring in PEEK posts did not cause tooth fracture, indicating suitability for re-posting. Özarslan et al. (2021) also stated that PEEK, glass fiber, and zirconium oxide post and cores in all groups showed sufficient fracture resistance to withstand maximum forces in the anterior region. Güven et al. (2020) compared the bond strengths of custom-made PEKK posts with prefabricated fiber and prefabricated PEKK posts. They emphasized that custom-made PEKK posts showed the highest bond strength in the cervical section, while fiber posts showed better bond strength in the middle and apical sections. The study demonstrated that custom-made PEKK posts could be an alternative to fiber posts.'

Applications of PEEK and PEKK in Implantology:

PEEK and PEKK materials have demonstrated considerable potential in dental implantology. They are utilised to fabricate healing abutments, implant abutments and implant-supported prostheses due to their excellent mechanical properties and biocompatibility. Recent research has explored the long-term success and stability of PEEK and PEKK frameworks for implant-supported prostheses, emphasising their clinical effectiveness and patient satisfaction.

Schwitalla et al., in their study, compared PEEK and titanium abutments with each other and compared the stress values occurring around the bone. In this study, the stress created by PEEK was lower (Schwitalla et al., 2015). A study observed that the force transmitted to the surrounding tooth tissues and cementum was minimised due to the low elastic modulus of abutments made with PEEK material (Schwitalla AD et al. 2016, ÇALIŞKAN, C., 2020). In the study by Val et al. in 2016, They stated that titanium-reinforced PEEK abutments could be an effective alternative to conventional abutments, considering their high biocompatibility, and they will ensure the stability of the bone level and soft tissue (Mate Sanchez et al., 2016). Considering the biocompatibility of the PEEK material, the healing abutment can also be produced from PEEK. Koutouzis et al., as a result of a randomised controlled clinical trial performed, it was reported that there was no significant difference between soft tissue inflammation and bone resorption around PEEK and titanium abutments (Koutouzis et al., 2011).

In a study by Sirandoni et al. in 2019, they evaluated the biomechanical behaviour of different framework materials in implant-supported fixed mandibular prostheses. Six groups were evaluated based on the framework material: titanium, cobalt-chromium, zirconia, PEEK, carbon fiber-reinforced PEEK (CFR-PEEK), and PMMA. PEEK and PMMA frameworks showed decreases in von Mises stresses in implants, abutments, and frameworks while exhibiting the highest total deformation values with high tensile stresses reaching critical values in trabecular bone. In a study conducted by Dayan et al. in 2021, they examined the effect of framework material on stress distribution in maxillary full-arch fixed prostheses supported by four implants. They evaluated Poly-Ether-Ketone-Ketone (PEKK), PEEK, titanium, and monolithic zirconia framework materials. PEKK and PEEK transmitted less stress to the cortical bone, and considering the current results of PEKK and PEEK materials, they found that compared to zirconia and titanium frameworks, they exhibited higher stress transmission to prosthetic components. They emphasised that clinicians should be aware of the possibility of screw loosening and fracture when using PEKK or PEEK in All-on-Four frameworks (Dayan & Geckili, 2021).

Applications of PEEK and PEKK in Orthodontics:

In orthodontics, PEEK and PEKK materials are used to fabricate clear aligners, orthodontic appliances, and temporary anchorage devices (TADs). These materials offer excellent clarity, flexibility, and biocompatibility, providing esthetic and comfortable orthodontic solutions. Recent research has focused on optimizing PEEK and PEKK orthodontic appliance design and fabrication to enhance treatment outcomes (Maekawa et al., 2015)

As a result of the research on the effects of standard metal brackets and PEEK bracket wires, PEEK bracket wires have less friction resistance, higher chemical resistance, and higher

abrasion resistance, which are the prominent advantages over metal bracket wires (Lesti et al. 2017).

CONCLUSION

High-performance polymers are materials of increasing interest in dentistry. PEEK and PEKK materials have emerged as versatile and promising materials in various dental disciplines. Their exceptional properties, including mechanical strength, biocompatibility, and clinical versatility, have expanded treatment options and improved clinical outcomes. It is used especially in implants, prosthetic restorations, and other dentistry fields such as orthodontics and endodontics. Due to their superior mechanical properties, high-performance polymers will likely be used as an alternative to titanium and its alloys. Ongoing research and advancements in manufacturing techniques, surface modifications, and customisation are expected to improve further the utilisation of PEEK and PEKK materials in dentistry, contributing to the advancement of dental materials and patient care. Long-term clinical studies on the use of the material are limited. For this reason, using the material in dental practice has not yet gained sufficient momentum. More studies are needed to be an alternative to long-existing metals.

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